

BREAK INTO AEROSPACE

Electric & Hybrid-Electric Aviation

Subdomain Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

Electric and hybrid-electric aviation is the fastest-restructuring subdomain in aerospace engineering. The sector has rapidly transitioned from academic research to certificated commercial aircraft types, trillions in investment, and the publication of dedicated regulatory frameworks like EASA SC-VTOL-01. The core challenge lies in packing energy into flying structures, converting it through high-efficiency electric machines, and managing thermal consequences within razor-thin weight margins.

The field sits at the critical intersection of electrochemistry, power electronics, motor design, and airworthiness. The employer landscape spans pure eVTOL developers (Joby, Archer, Beta), hybrid regional aircraft (Heart Aerospace), and established OEMs (Airbus, Boeing/Wisk, Embraer/Eve). In a sector where handbook methods do not yet exist, success depends on the ability to move fluently across technical interfaces.

PART I · WHAT ELECTRIC & HYBRID AVIATION ACTUALLY LOOKS FOR

By Level

- **Intern / Graduate** — Employers screen for physical understanding of energy conversion and the ability to build a weight-and-power budget without a safety net of established methods. *Distinguishing Question:* “A 4-passenger eVTOL needs 120 kW for 15 minutes. Estimate battery mass at 150, 200, and 250 Wh/kg specific energy. What does this tell you about mission viability?” Candidates must work in SI units, stating depth-of-discharge and reserve assumptions clearly.
- **Mid-Level (Engineer, Senior Engineer, ~3–8 years)** — Interviews focus on solving specific startup-culture design problems: sizing a motor for torque/thermal constraints or assessing if an inverter topology meets switching frequency needs. *Current Context:* Major hiring is driven by FAA G-1 issue papers and SC-VTOL-01 safety requirements. Engineers who arrive having studied public certification filings from Joby or Archer are highly distinguished.
- **Senior (Principal Engineer, Technical Lead, Chief Engineer)** — At this level, the focus is architectural decision-making under deep technical uncertainty. You must be technically rigorous regarding electrochemical degradation while comfortable negotiating certification bases with authorities for technologies that lack decades of in-service failure data.

Hands-On Experience and Domain Knowledge Treated as Strong Signals

Domain	Why It Signals Depth
Li-ion Electrochemistry	Understanding intercalation, SEI formation, and C-rate relationships. Knowing why 4C discharge produces nonlinear heating compared to 1C.
BMS Design	State-of-Charge (SOC) and State-of-Health (SOH) estimation using Kalman filters. Designing architectures where no single point of failure results in loss of propulsion.
Electric Machine Design	Proficiency in PMSM (Permanent Magnet Synchronous Motors). Understanding Lorentz force torque, field weakening for high-speed operation, and thermal insulation limits.
Power Electronics	Three-phase inverter topologies using Silicon Carbide (SiC) MOSFETs. Understanding switching losses and the EMI challenges created by high-frequency power conversion.
Thermal Management	Designing liquid cooling or phase-change material systems to reject heat from battery internal resistance and iron/copper losses during takeoff.
SC-VTOL-01 / FAA Powered-Lift	Familiarity with EASA's Special Condition or the FAA's new powered-lift category. Knowing how propulsion system integrity demonstration differs from CS-25.
Distributed Electric Propulsion	Leveraging multiple propulsors for boundary layer ingestion (BLI) and using motor speed differentials as flight control effectors.
Energy vs. Power Density	Navigating the Ragone plot. Selecting chemistries (NMC, LFP, solid-state) based on whether the mission is energy-limited (cruise) or power-limited (hover).
Hydrogen Fuel Cells (PEMFC)	Understanding balance-of-plant weight penalties, hydrogen embrittlement, and the efficiency limits of electrochemical oxidation for zero-emission flight.

How Electric Aviation Interviews Differ from Generic Roles

- **The Weight Budget is the Sovereign Constraint.** Unlike automotive or consumer electronics, a 50 kg increase in battery mass is often disqualifying for an eVTOL payload. Every technical choice is evaluated first for its mass consequence.
- **Certification is Developed, Not Just Applied.** There is no 70-year playbook. Senior engineers are expected to help develop means of compliance (MOC) for novel failure modes like battery thermal runaway containment.
- **Novel Failure Modes Require New Methodologies.** Traditional FTAs and FMEAs must be adapted for SiC MOSFET shoot-through, common-mode voltage damage, and cell-level thermal runaway propagation.
- **Hardware-in-the-Loop (HIL) is the Development Bed.** In the absence of a large fleet, reliability cases are built from accelerated life testing (ALT) and high-fidelity physics-based degradation models.

PART II · 20 SPECIFIC PREPARATION TASKS

Intern / Graduate

1. Build a complete electric propulsion system energy budget for a specified eVTOL mission 5–6 h

Skill: Systems-level quantitative energy reasoning. **Objective:** Calculate required battery mass from phase-specific power (hover, transition, cruise), durations, conversion losses, and 20% reserves. **Done Signal:** You can state the battery mass fraction for a 4-pax, 15-minute mission and explain why a specific energy of 150 Wh/kg makes most eVTOL missions payload-impossible.

2. Understand Li-ion fundamentals — electrochemistry, C-rate, and the thermal runaway cascade 5–6 h

Skill: Electrochemical literacy. **Objective:** Learn why capacity fades at low temps (diffusion limits) and what triggers the NMC decomposition cascade at 200°C. **Done Signal:** You can sketch a discharge voltage curve and explain what the SEI (Solid Electrolyte Interphase) does for cell aging.

3. Understand electric motor fundamentals — PMSM principles and the torque-speed efficiency map 4–5 h

Skill: Electric machine physics. **Objective:** Learn about back-EMF (K_e), field-weakening current, and iron vs. copper losses. **Done Signal:** You can sketch a PMSM efficiency map, identifying the “high-efficiency island” and the thermal limits that govern continuous power.

4. Prepare behavioural answers demonstrating systems thinking and comfort with uncertainty 2–3 h

Skill: Startup-culture adaptability. **Objective:** Focus on a time you built an analytical method from first principles because no standard existed. **Done Signal:** You have stories ready involving real electrical hardware or data where you made a decision based on incomplete information.

5. Understand power electronics — inverter topology, SiC vs. IGBT, and switching losses 5–6 h

Skill: High-voltage power conversion. **Objective:** Draw a three-phase two-level VSI. Explain why SiC enables smaller filter inductors. **Done Signal:** You can explain the switching loss mechanism (voltage-current overlap) and its thermal management implications.

6. Get hands-on with battery cell characterisation — internal resistance and fade 6–10 h

Skill: Empirical electrochemical measurement. **Objective:** Perform CC-CV charging on a cell and measure internal resistance via pulse technique. **Done Signal:** You can plot capacity retention vs. cycle number and explain how C-rate affects the SEI growth rate.

Mid-Level (~3–8 years)

7. Build thorough working knowledge of BMS design — state estimation and airworthiness safety **Skill:** Safety-critical electronic control. **Objective:** Learn about Kalman-filter SOC estimation and redundant measurement paths required for SC-VTOL-01. **Done Signal:** You can describe the contactor control logic (including pre-charge) and why 5% SOC error is catastrophic for reserves. 5–6 h

8. Understand the SC-VTOL-01 framework and current ELOS challenges **Skill:** Regulatory framework competency. **Objective:** Study Category Enhanced (CAT ENH) requirements for loss of a single propulsor. **Done Signal:** You can explain what an Equivalent Level of Safety (ELOS) finding is and why it's used for novel battery chemistries. 5–6 h

9. Prepare a story about diagnosing a battery or motor system problem using test data **Skill:** Applied electric propulsion engineering. **Objective:** Focus on a time hardware behaved differently than the simulation (e.g., higher thermal resistance than data-sheet). **Done Signal:** Your story includes specific numerical values (predicted vs. measured) and a clear engineering lesson learned. 2 h

10. Understand thermal runaway propagation — containment, testing, and guidance **Skill:** High-stakes battery safety. **Objective:** Learn about intumescent barriers and pressure relief venting required to pass DO-311A tests. **Done Signal:** You can describe the cell-to-cell propagation mechanism and the EASA guidance for showing non-propagation. 5–6 h

11. Understand distributed electric propulsion (DEP) control and motor failure response **Skill:** Propulsion-flight control integration. **Objective:** Learn how control allocation algorithms redistribute torque when 1 of 8 motors fails. **Done Signal:** You can explain the attitude control authority envelope after a motor loss in the hover regime. 4–5 h

12. Build working knowledge of hover aerodynamics and the power-required curve **Skill:** Rotor-lift physics. **Objective:** Derive hover power from actuator disk theory: $P \propto (W/n)^{1.5} / \sqrt{\rho A}$. **Done Signal:** You can sketch the power-required curve for a tiltrotor, identifying the “transition power hump.” 3–4 h

13. Prepare for a power system sizing problem under interview conditions **Skill:** Structured design speed. **Objective:** Practice sizing a battery and motor system for a specific mission (e.g., 2-seater trainer) in <45 mins. **Done Signal:** You have practiced this twice, including motor overload factor 4–6 h

and thermal management weight penalty estimates.

4–6 h

Senior Level (*Principal Engineer, Technical Lead, Chief*)

14. Prepare to walk through the complete development of an electric component **Skill:** End-to-end certification ownership. **Objective:** Trace a pack or motor from system safety requirements through qualification evidence. **Done Signal:** You can discuss how you negotiated a novel means of compliance with EASA or the FAA for a specific hardware finding.

4–6 h

15. Prepare a technically grounded position on an active electric aviation frontier **Skill:** Forward-looking technical leadership. **Objective:** Analyze ionic conductivity limits in solid electrolytes or rotor integrity at 20,000 RPM. **Done Signal:** You have a 3-minute talk ready that identifies a specific *unsolved* engineering problem (not just a generic opportunity).

4–5 h

16. Lead a discussion on architecture trades: series vs. parallel vs. pure electric **Skill:** Architectural speed and reasoning. **Objective:** Compare architectures for a 19-pax commuter in terms of energy chain and infrastructure dependency. **Done Signal:** You can recommend a specific architecture for a 200 nm mission, stating sensitivities to technology assumptions.

4–5 h

17. Understand the FAA powered-lift pathway and Part 135 rulemaking **Skill:** US regulatory strategy. **Objective:** Learn how the FAA's powered-lift aircraft category differs from small rotorcraft (CS-27). **Done Signal:** You can describe two areas where the FAA and EASA diverge in propulsion safety requirements.

4–5 h

18. Build fluency in estimating power, energy, and thermal loads without simulation **Skill:** Senior-speed sanity checking. **Objective:** Estimate hover power, iron losses, and switching losses using first principles. **Done Signal:** You have worked through 5 problems (e.g., coolant flow rate from heat rejection) out loud, stating assumptions first.

4–5 h

All Levels

19. Prepare a specific answer to "Why electric aviation?" reflecting genuine technical engagement **Skill:** Intellectual character signal. **Objective:** Name a specific challenge (e.g., the H2 balance-of-plant weight problem) that excites you. **Done Signal:** You have a 90-second answer that proves you are a specialist, not just a generic sustainable-aviation fan.

1–2 h

20. Map your background to a specific employer type and development stage **Skill:** Precise self-positioning. **Objective:** Understand if you belong in a development startup (Joby/Archer) or a production ramp-up role. **Done Signal:** You can state your tech domain and how it contributes to the employer's current certification phase in <60s. 1–2 h

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	Mission energy budget — battery mass estimation	All	5–6 h
2	Li-ion cell fundamentals and thermal runaway	All	5–6 h
3	PMSM fundamentals and efficiency maps	All	4–5 h
4	STAR stories — technical uncertainty	All	2–3 h
5	Power electronics — SiC inverters and losses	All	5–6 h
6	Practical cell characterisation cycling	All	6–10 h
7	BMS design and safety architecture	Mid+	5–6 h
8	SC-VTOL-01 framework and ELOS cases	Mid+	5–6 h
9	Hardware discrepancy diagnosis story	Mid+	2 h
10	Thermal runaway propagation management	Mid+	5–6 h
11	DEP control allocation and failure response	Mid+	4–5 h
12	eVTOL aerodynamics and power curves	Mid+	3–4 h
13	Propulsion design problem (timed)	Mid+	4–6 h
14	Complete propulsion certification story	Senior	4–6 h
15	Active frontier position (Solid-state/Hydrogen)	Senior	4–5 h
16	Architecture trade-off live discussion	Senior	4–5 h
17	FAA powered-lift certification pathway	Senior	4–5 h
18	Quantitative estimation fluency — without sim	Senior	4–5 h
19	Technically specific “Why electric?” answer	All	1–2 h
20	Employer and development stage positioning	All	1–2 h

THE STRUCTURAL REALITY

Electric and hybrid aviation is the only aerospace subdomain where critical engineering challenges are simultaneously electrochemical, regulatory, and commercial. The energy density required for regional flight doesn't yet exist at manufacturable cost, and the regulations to certify novel chemistries have no precedent.

This circularity creates a unique environment. Engineers are simultaneously designing hardware, developing qualification tests, and negotiating standards. Every assumption has larger uncertainty than in any mature aerospace discipline.

Successful engineers are rigorous about the physics they can characterise (thermodynamics) and humble about what they cannot (long-term degradation). They use data aggressively where it exists and conservatively where it doesn't. In a sector where a single high-profile failure could set back entire trillions in investment, the conservative decision is almost always the right one.

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